

Erdős Problem #954

EP-954

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $1 = a_1 < a_2 < \dots$ be the sequence of integers defined by $a_1 = 1$ and a_{k+1} is the smallest integer n for which the number of solutions to $a_i + a_j \leq n$ (with $i \leq j \leq k$) is less than $n - k$.

Is the number of solutions to $a_i + a_j \leq x$ equal to $x + O(x^{1/4+o(1)})$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>